



Layer Normalization

Jimmy Lei Ba, Jamie Ryan Kiros, Geoffrey E. Hinton

[arXiv:1607.06450](https://arxiv.org/abs/1607.06450)

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Eunjae Son

MINDS Lab

- ☐ **Introduction**
- ☐ **Batch Normalization**
 - In CNN
 - In RNN
- ☐ **Layer Normalization**
- ☐ **Conclusion**

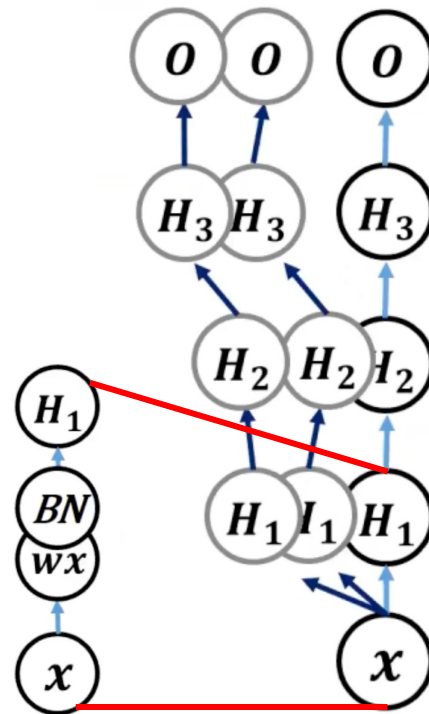
Introduction

□ Challenge of Training Deep Neural Networks

- Long training time

□ Batch Normalization

- To reduce training time by reducing **Internal Covariate Shift**
- Including additional normalization stages in DNN
- Using mean and standard deviation **across the training data**



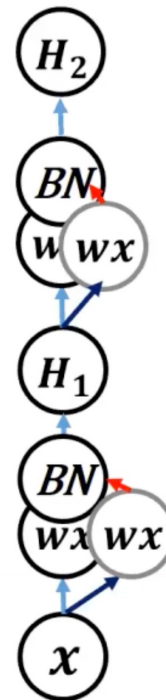
Introduction

□ Challenge of Training Deep Neural Networks

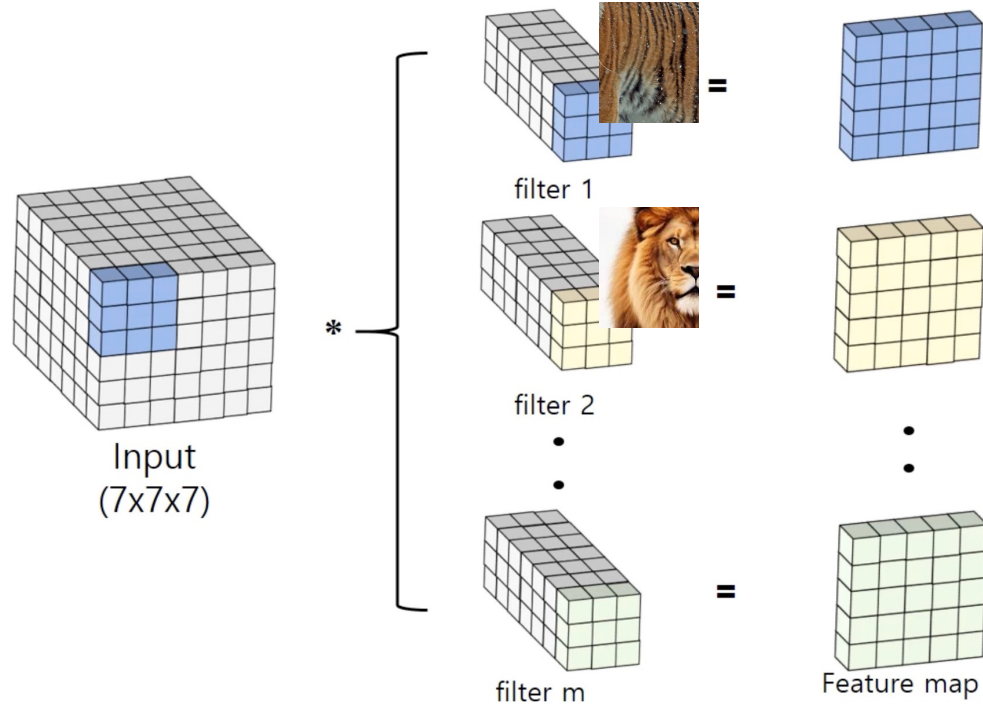
- Long training time

□ Batch Normalization

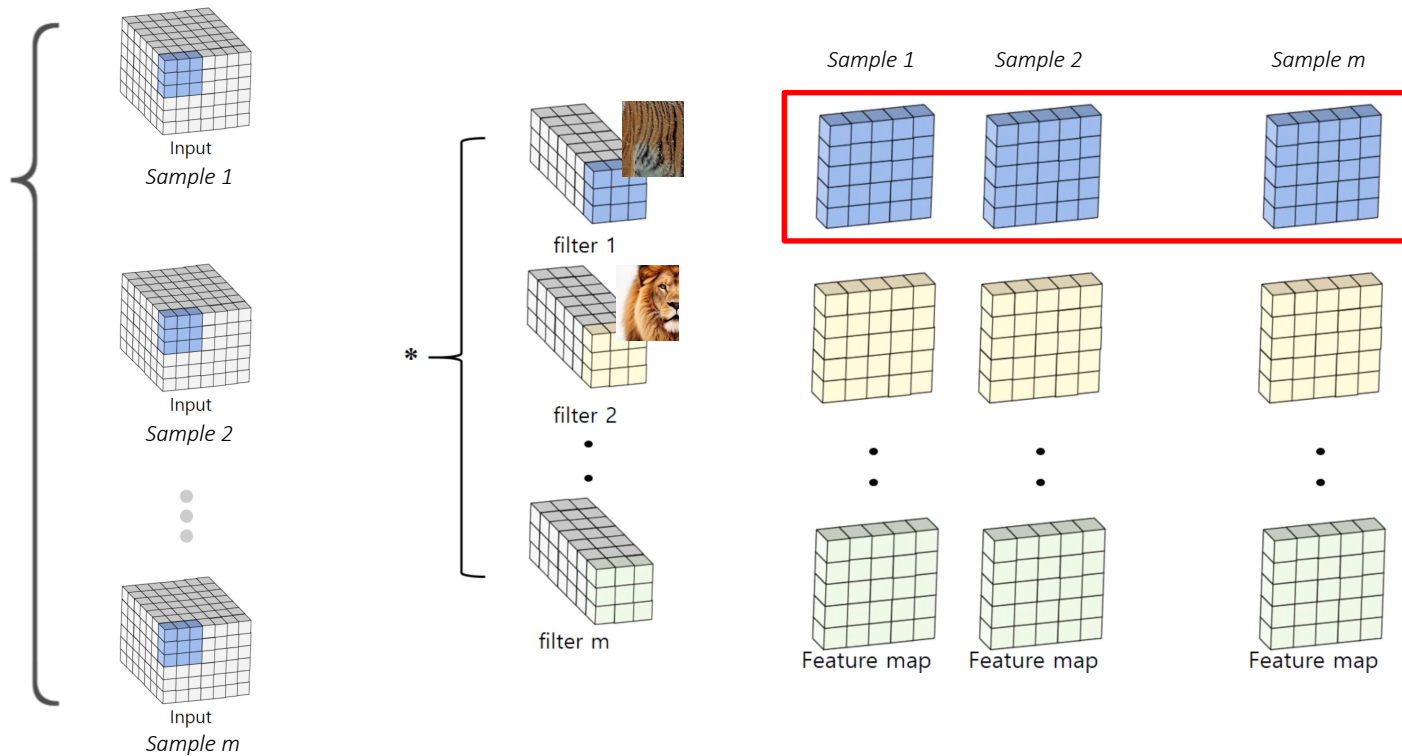
- To reduce training time by reducing **Internal Covariate Shift**
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Batch Normalization: CNN



Batch Normalization: CNN

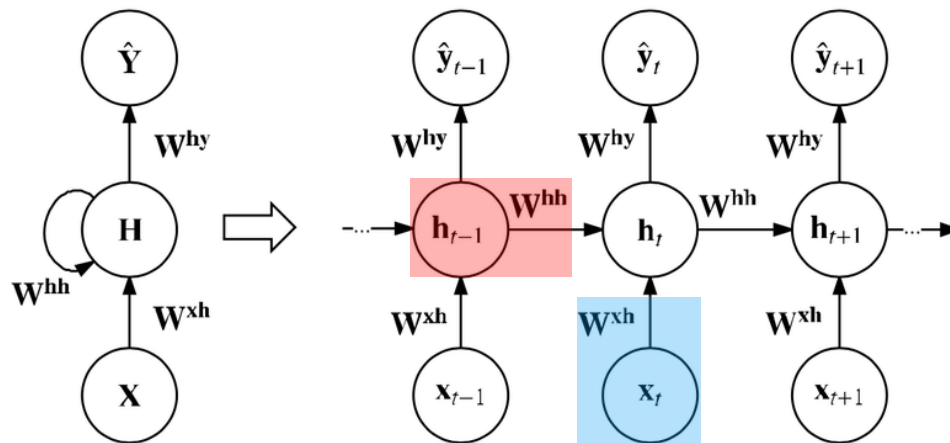


Background: RNN

□ Recurrent Neural Network

■ For processing sequential data

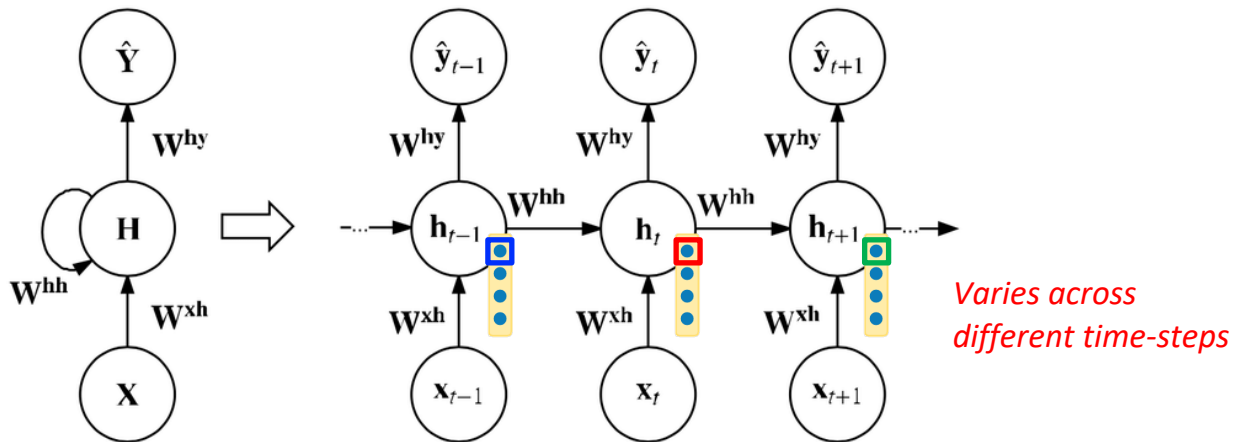
■ Summed inputs in the recurrent layer $\mathbf{a}^t = \mathbf{W}_{hh}\mathbf{h}^{t-1} + \mathbf{W}_{xh}\mathbf{x}^t$



□ Recurrent Neural Network

A | 그날 아침 마신 커피로 인해 식중독에 걸려 병원에 갔고, ...

B | 그날 저녁 방화 사건으로 훼손된 문화재를 복원하기 위해 기술 Y가 사용되었고, ...

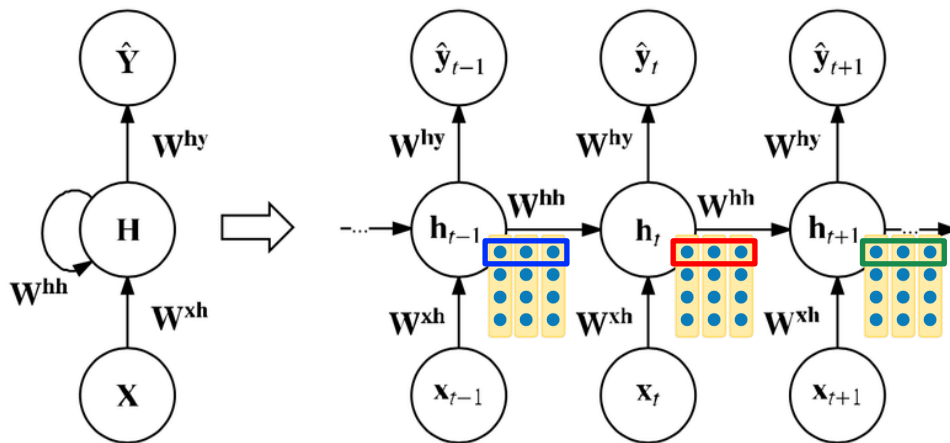


Batch Normalization: RNN ?

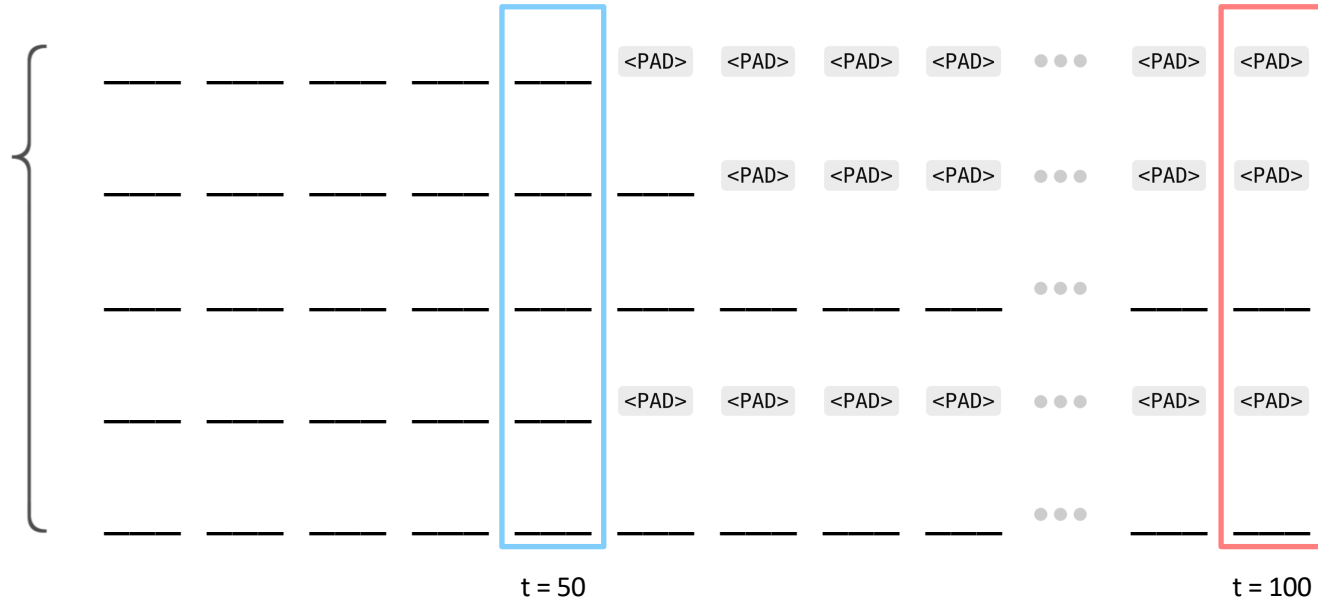
□ Not obvious how to apply to RNN

- Batch Normalization depends on Batch Statistics

- estimate of the whole training data distribution



□ Limitation 1: Unstable Statistics with Variable Sequence Lengths

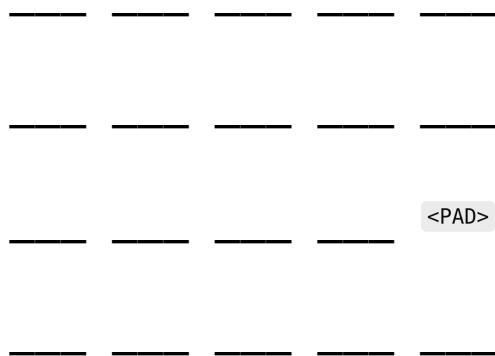


Batch Normalization: RNN ?

□ Limitation 2: Requires Running Averages

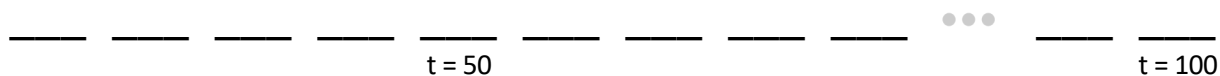
- Problematic if a test sequence is longer than any of the training sequences

Training



*Moving averages unavailable
from t=51 onward*

Test



Batch Normalization: RNN ?

□ Limitation 2: Requires Running Averages

■ Problematic even if ...



Batch Normalization: RNN ?

□ Fundamental Mismatch between BN and RNN

I like apple

Trump wants the GOP to nationalize US elections

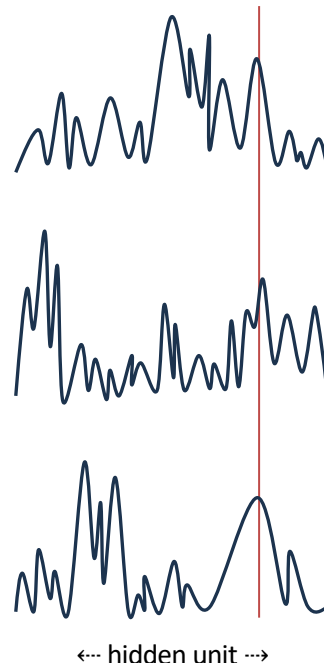
In quantum mechanics, ...

⋮

Batch Normalization: RNN ?

□ Fundamental Mismatch between BN and RNN

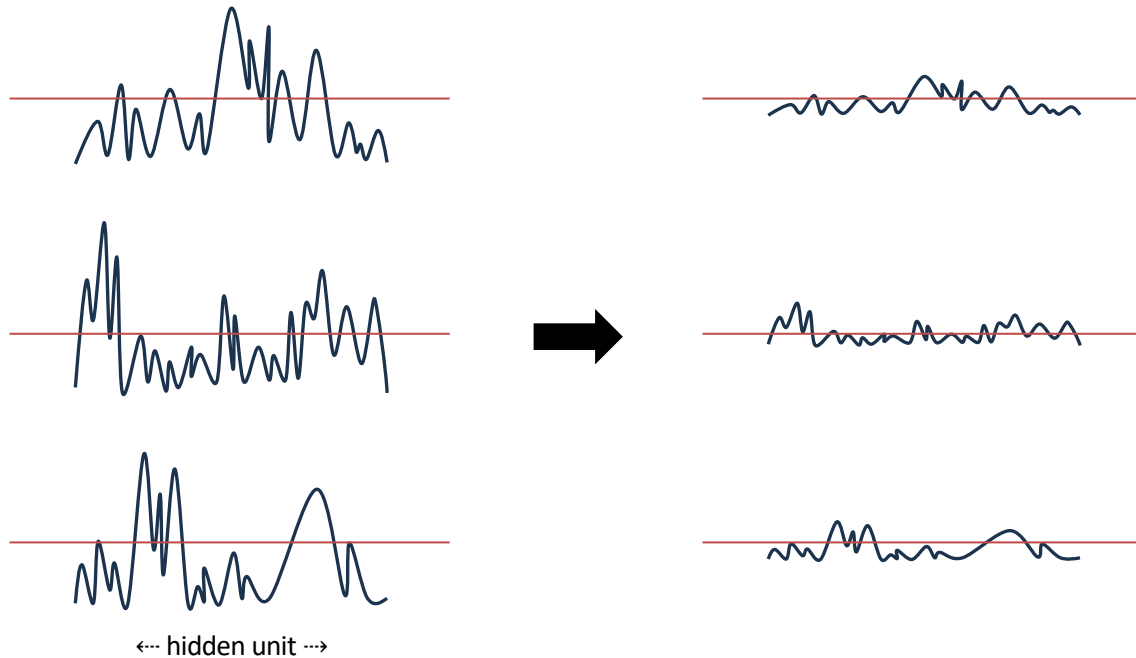
{
I like apple
Trump wants the
In quantum mechanics
t = 3



Layer Normalization

□ Idea

- Computing mean and variance on a **single** training case (within a hidden layer)

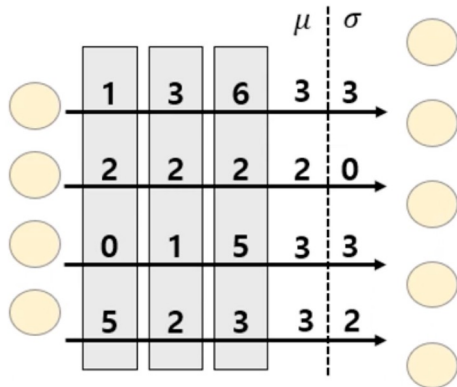


Layer Normalization

□ Idea

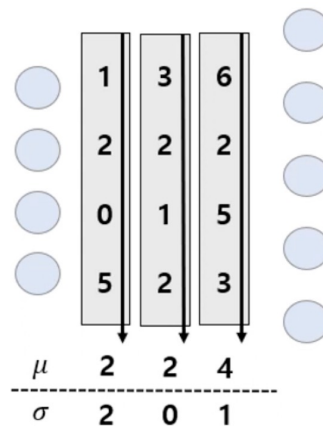
- Computing mean and variance on a **single** training case (within a hidden layer)

Batch Normalization



$$\mu_i^l = \mathbb{E}_{\mathbf{x} \sim P(\mathbf{x})} [a_i^l] \quad \sigma_i^l = \sqrt{\mathbb{E}_{\mathbf{x} \sim P(\mathbf{x})} [(a_i^l - \mu_i^l)^2]}$$

Layer Normalization



$$\mu^l = \frac{1}{H} \sum_{i=1}^H a_i^l \quad \sigma^l = \sqrt{\frac{1}{H} \sum_{i=1}^H (a_i^l - \mu^l)^2}$$

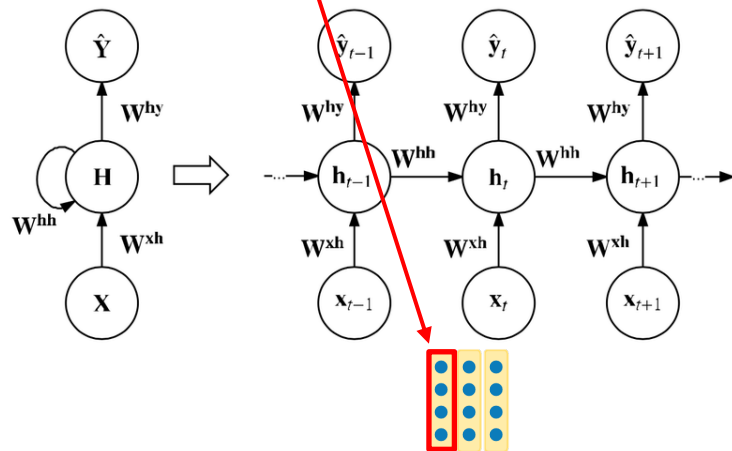
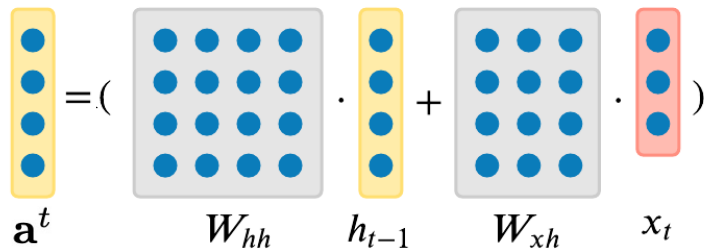
Layer Normalization

□ Method

■ Apply to RNN $\mathbf{h}^t = f \left[\frac{\mathbf{g}}{\sigma^t} \odot (\mathbf{a}^t - \mu^t) + \mathbf{b} \right]$

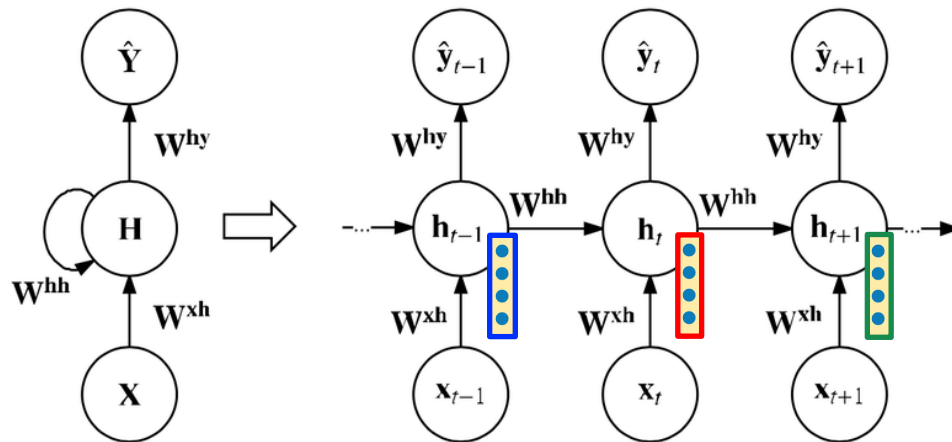
$$\mu^t = \frac{1}{H} \sum_{i=1}^H a_i^t \quad \sigma^t = \sqrt{\frac{1}{H} \sum_{i=1}^H (a_i^t - \mu^t)^2}$$

$$\mathbf{a}^t = W_{hh} \mathbf{h}^{t-1} + W_{xh} \mathbf{x}^t$$



□ Layer Normalized RNN

- Standard RNN, average magnitude of summed inputs grow/shrink at every time-step
Leading to exploding or vanishing gradients
- LN $\Rightarrow$ Invariant to re-scaling, More stable hidden-to-hidden dynamics



□ 6.2 Teaching machines to read and comprehend

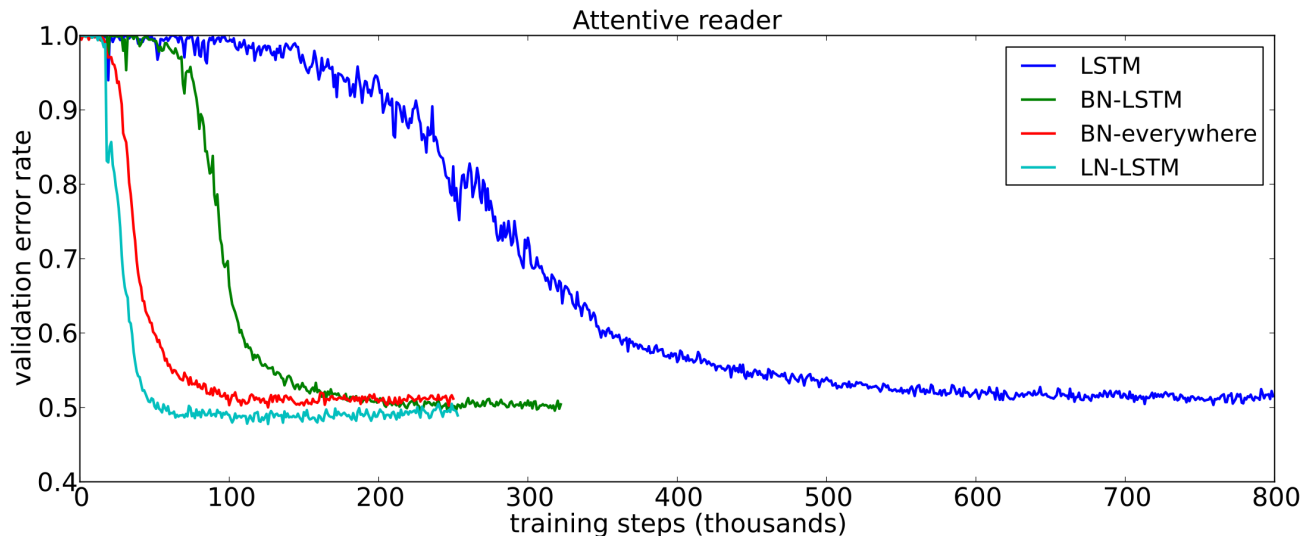
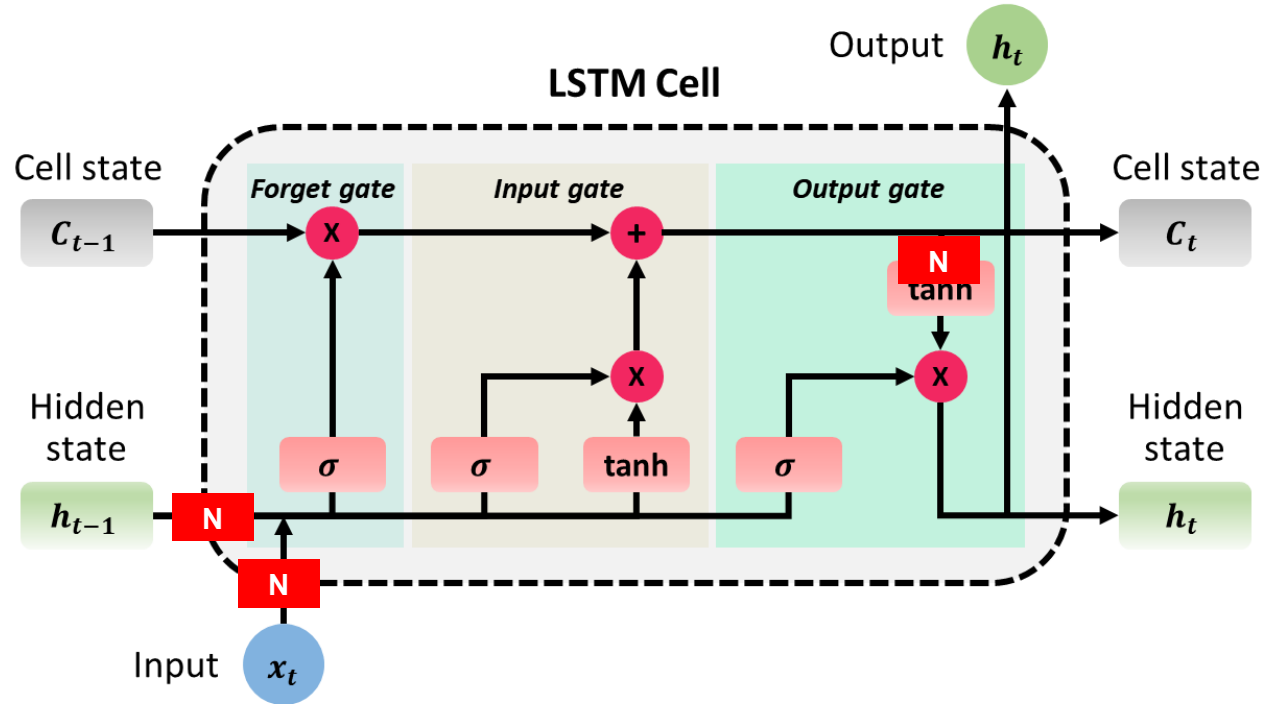


Figure 2: Validation curves for the attentive reader model. BN results are taken from [Cooijmans et al., 2016].

Layer Normalization: Experiments

□ 6.2 Teaching machines to read and comprehend



□ 6.5 Handwriting sequence generation

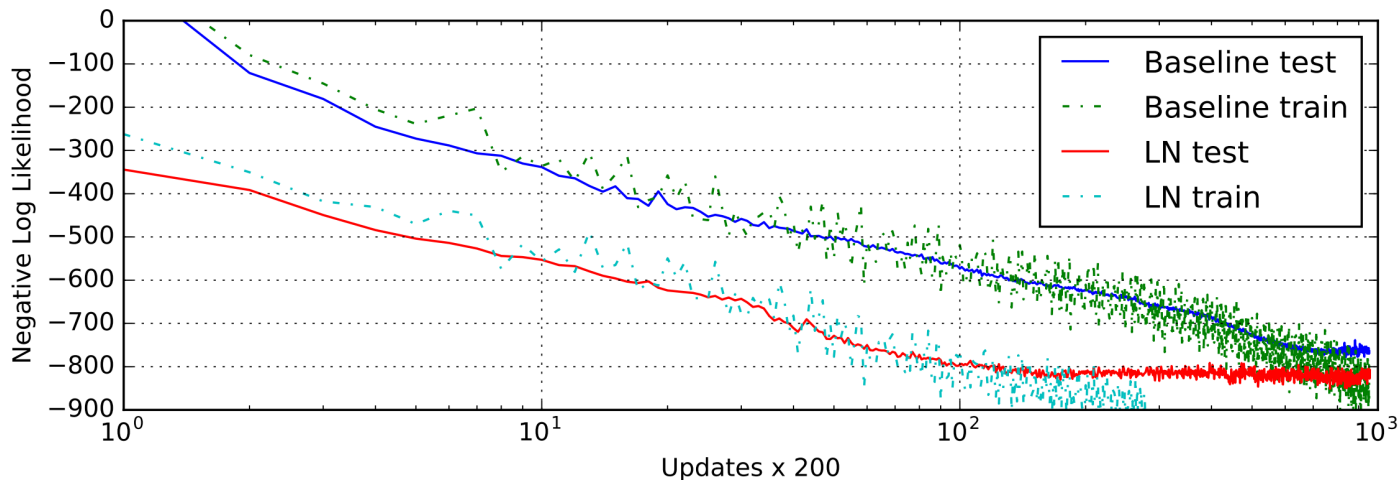
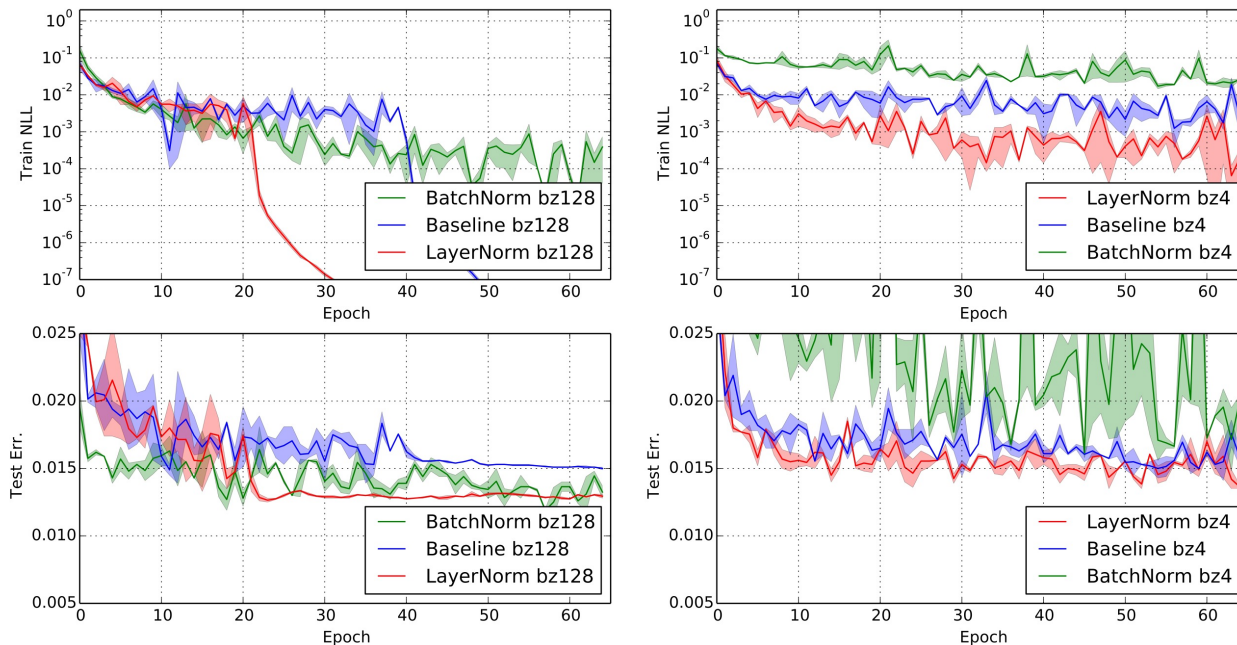


Figure 5: Handwriting sequence generation model negative log likelihood with and without layer normalization. The models are trained with mini-batch size of 8 and sequence length of 500,

Layer Normalization: Experiments

□ 6.6 Permutation invariant MNIST (Figure 6)



Conclusion

- ☐ **Layer Normalization: Sample-wise**
- ☐ Exactly same computation at training and test times (unlike BN)
- ☐ Apply to RNN - Stabilize hidden state dynamics
- ☐ Reduce training time
- ☐ Especially effective for long sequences and small mini-batches